

Maths - Question 48

Original paper page 46; original question and option regions

If the equation of the polar of the point $(\alpha, -1)$ with respect to the circle

$$x^2 + y^2 - 4x - 6y - 12 = 0 \text{ is } y = \beta, \text{ then } 4(\alpha + \beta) =$$

$x^2 + y^2 - 4x - 6y - 12 = 0$ వృత్తం దృష్ట్యా $(\alpha, -1)$ బిందువు యొక్క ధ్రువరేఖ సమీకరణం $y = \beta$ అయితే, $4(\alpha + \beta) =$

Options :

1. ✓ -5

2. ✗ 7

3. ✗ -6

4. ✗ 0

Maths - Question 49

Original paper page 47; original question and option regions

If θ is the angle between the tangents drawn from the point $(-1, -1)$ to the circle $x^2 + y^2 - 4x - 6y + c = 0$ and $\cos \theta = -\frac{7}{25}$, then the radius of the circle is

$x^2 + y^2 - 4x - 6y + c = 0$ వృత్తానికి $(-1, -1)$ బిందువు నుండి గీచిన స్పర్శరేఖల మధ్య గల కోణం θ అవుతూ మరియు $\cos \theta = -\frac{7}{25}$ అయితే, ఆ వృత్తం యొక్క వ్యాసార్థం

Options :

1. ✓ 4
2. ✗ 1
3. ✗ 2
4. ✗ 3

Maths - Question 50

Original paper page 47; original question and option regions

If the power of the point $(1, 6)$ with respect to the circle $x^2 + y^2 + 4x - 6y - a = 0$ is -16 , then $a =$

వృత్తం $x^2 + y^2 + 4x - 6y - a = 0$ దృష్ట్యా బిందువు $(1, 6)$ యొక్క బిందు శక్తి -16 అయితే, $a =$

Maths - Question 50 (continued)

Original paper page 48; original question and option regions

7

11

13

21

Maths - Question 51

Original paper page 48; original question and option regions

The radius of the circle passing through the points of intersection of the circles $x^2 + y^2 + 2x + 4y + 1 = 0$, $x^2 + y^2 - 2x - 4y - 4 = 0$ and intersecting the circle $x^2 + y^2 = 6$ orthogonally is

$x^2 + y^2 + 2x + 4y + 1 = 0$, $x^2 + y^2 - 2x - 4y - 4 = 0$ వృత్తాల ఖండన బిందువుల గుండాపోతూ, $x^2 + y^2 = 6$ వృత్తాన్ని లంబచ్ఛేదనం చేసే వృత్తము యొక్క వ్యాసార్థం

Options :

$$\sqrt{19}$$

1. ✖

Maths - Question 51 (continued)

Original paper page 49; original question and option regions

5

$$\sqrt{39}$$

4

Maths - Question 52

Original paper page 49; original question and option regions

The lengths of the two focal chords of the parabola $y^2 = 16x$ is 25 units each. If these two chords cut the parabola at A, B, C and D, then the area (in sq. units) of the quadrilateral formed by A, B, C and D is

$y^2 = 16x$ పరావలయమునకు గల రెండు నాభిజ్యాలలో ఒక్కొక్క దాని పొడవు 25 యూనిట్లు. ఈ రెండు జ్యాలు ఆ పరావలయాన్ని A, B, C మరియు D ల వద్ద ఖండిస్తే, A, B, C మరియు D లచే ఏర్పడే చతుర్భుజ వైశాల్యం (చ॥ యూనిట్లలో)

Options :

1. ✖ $\frac{625}{2}$

2. ✖ 180

Maths - Question 52 (continued)

Original paper page 50; original question and option regions

150

300

Maths - Question 53

Original paper page 50; original question and option regions

If the tangents drawn from a point P to the ellipse $4x^2 + 9y^2 - 16x + 54y + 61 = 0$ are perpendicular, then the locus of P is

P అనే ఒక బిందువు నుండి $4x^2 + 9y^2 - 16x + 54y + 61 = 0$ దీర్ఘ వృత్తమునకు గీచిన స్పర్శరేఖలు లంబంగా ఉంటే, అప్పుడు P యొక్క బిందుపథం

Options :

1. ✖ $x^2 + y^2 - 4x + 6y + 4 = 0$

2. ✔ $x^2 + y^2 - 4x + 6y = 0$

3. ✖ $x^2 + y^2 - 6x + 4y + 9 = 0$

4. ✖ $x^2 + y^2 - 6x + 4y = 0$

Maths - Question 54

Original paper page 51; original question and option regions

$x + y + 3 = 0$, $2x - y + 1 = 0$ are the equations of the asymptotes of a hyperbola. If $(1, -2)$ is a point on this hyperbola, then the equation of its conjugate hyperbola is

$x + y + 3 = 0$, $2x - y + 1 = 0$ లు ఒక అతిపరావలయం యొక్క అనంత స్పర్శరేఖల సమీకరణాలు. ఈ అతిపరావలయం మీద $(1, -2)$ ఒక బిందువైతే, దీని సంయుగ్మ అతిపరావలయం యొక్క సమీకరణం.

Options :

1. ✖ $2x^2 + xy - y^2 + 7x - 2y - 1 = 0$

2. ✔ $2x^2 + xy - y^2 + 7x - 2y + 13 = 0$

3. ✖ $2x^2 + xy + y^2 - 7x - 2y - 1 = 0$

4. ✖ $2x^2 + xy + y^2 - 7x - 2y + 13 = 0$

Maths - Question 55

Original paper page 52; original question and option regions

If θ is the acute angle between the tangents drawn from the point $(1, 1)$ to the hyperbola $4x^2 - 5y^2 - 20 = 0$, then $\tan \theta =$

$4x^2 - 5y^2 - 20 = 0$ అతిపరావలయానికి $(1, 1)$ బిందువు నుండి గీచిన స్పర్శరేఖల మధ్య గల లఘుకోణం θ అయితే, అప్పుడు $\tan \theta =$

Options :

1. ✓ $2\sqrt{21}$

2. ✗ $\frac{4}{5}$

3. ✗ $\frac{\sqrt{7}}{2}$

4. ✗ $\frac{2}{\sqrt{7}}$

Maths - Question 58

Original paper page 55; original question and option regions

A plane π given by $ax + by + 11z + d = 0$ is perpendicular to the planes $2x - 3y + z = 4$, $3x + y - z = 5$ and the perpendicular distance from the origin to the plane π is $\sqrt{6}$ units. If all the intercepts made by the plane π on the coordinate axes are positive, then $d =$

$ax + by + 11z + d = 0$ గా ఇవ్వబడిన π అనే ఒక తలం, $2x - 3y + z = 4$, $3x + y - z = 5$ తలాలకు లంబంగా ఉంటుంది మరియు మూల బిందువు నుండి π తలానికి గల లంబదూరం $\sqrt{6}$ యూనిట్లు. π తలము నిరూపక అక్షాలపై ఏర్పరచే అంతర ఖండాలన్నీ ధనాత్మకము అయితే, అప్పుడు $d =$

Options :

1. ✖ ab
2. ✖ $-2ab$
3. ✖ $4ab$
4. ✔ $-3ab$

Maths - Question 59

Original paper page 56; original question and option regions

The quadratic equation whose roots are

$$l = \lim_{\theta \rightarrow 0} \left(\frac{3 \sin \theta - 4 \sin^3 \theta}{\theta} \right) \text{ and } m = \lim_{\theta \rightarrow 0} \left(\frac{2 \tan \theta}{\theta(1 - \tan^2 \theta)} \right) \text{ is}$$

$$l = \lim_{\theta \rightarrow 0} \left(\frac{3 \sin \theta - 4 \sin^3 \theta}{\theta} \right) \text{ మరియు } m = \lim_{\theta \rightarrow 0} \left(\frac{2 \tan \theta}{\theta(1 - \tan^2 \theta)} \right) \text{ లను మూలాలుగా గలిగిన}$$

వర్గ సమీకరణం

Options :

1. ✖ $x^2 + 5x + 6 = 0$

2. ✔ $x^2 - 5x + 6 = 0$

3. ✖ $x^2 - 5x - 6 = 0$

4. ✖ $x^2 + 5x - 6 = 0$

Maths - Question 60

Original paper page 57; original question and option regions

$$\lim_{x \rightarrow \infty} \frac{3x + 4\cos^2 x}{\sqrt{x^2 - 5\sin^2 x}} =$$

Options :

$$\frac{3}{5}$$

1. ✖

$$\frac{4}{5}$$

2. ✖

3. ✔ 3

4. ✖ 1

Maths - Question 61

Original paper page 58; original question and option regions

If a function $f(x) = \begin{cases} \frac{\sqrt[3]{1+ax^2+bx^3} - \sqrt[3]{1-ax^2-bx^3}}{x^2}, & x < 0 \\ 5, & x = 0 \\ \frac{\tan 3x - \sin 3x}{bx^3}, & x > 0 \end{cases}$

is continuous at $x = 0$, then the geometric mean of a and b is

$$f(x) = \begin{cases} \frac{\sqrt[3]{1+ax^2+bx^3} - \sqrt[3]{1-ax^2-bx^3}}{x^2}, & x < 0 \\ 5, & x = 0 \\ \frac{\tan 3x - \sin 3x}{bx^3}, & x > 0 \end{cases}$$

అనే ప్రమేయం $x = 0$ వద్ద అవిచ్ఛిన్నము అయితే, a మరియు b ల గుణ మధ్యమం

Options :

$$\frac{3}{2}$$

1. ✖

Maths - Question 61 (continued)

Original paper page 59; original question and option regions

$$\frac{9}{2}$$

$$\frac{81}{4}$$

$$\frac{9}{4}$$

Maths - Question 62

Original paper page 59; original question and option regions

If $y = \log(\sec(\tan^{-1}x))(x > 0)$, then $\frac{dy}{dx}$ at $x = 1$ is

$y = \log(\sec(\tan^{-1}x))(x > 0)$ అయితే, $x = 1$ వద్ద $\frac{dy}{dx} =$

Options :

1. ✖ 1

3

2. ✖

Maths - Question 62 (continued)

Original paper page 60; original question and option regions

$$\frac{1}{2}$$

$$\frac{3}{2}$$

Maths - Question 63

Original paper page 60; original question and option regions

If $y = \sin^{-1} \left(\frac{\sqrt{1 + \sin x} + \sqrt{1 - \sin x}}{\sqrt{1 + \sin x} - \sqrt{1 - \sin x}} \right)$ and $-\frac{3\pi}{2} < x < -\frac{\pi}{2}$, then $\frac{dy}{dx} =$

$y = \sin^{-1} \left(\frac{\sqrt{1 + \sin x} + \sqrt{1 - \sin x}}{\sqrt{1 + \sin x} - \sqrt{1 - \sin x}} \right)$ మరియు $-\frac{3\pi}{2} < x < -\frac{\pi}{2}$ అయితే, $\frac{dy}{dx} =$

Options :

$$-\frac{|\operatorname{cosec} \frac{x}{2}|}{2\sqrt{\sin^2 \frac{x}{2} - \cos^2 \frac{x}{2}}}$$

1. ✖

$$\frac{|\sec \frac{x}{2}|}{2\sqrt{\cos x}}$$

2. ✔

Maths - Question 63 (continued)

Original paper page 61; original question and option regions

$$\frac{\left| \cos \frac{x}{2} \right|}{2\sqrt{\cos x}}$$

$$\frac{\cos \frac{x}{2}}{\sqrt{\cos x}}$$

Maths - Question 64

Original paper page 61; original question and option regions

If $x = \sqrt{2} e^t (\sin t - \cos t)$ and $y = \sqrt{2} e^t (\sin t + \cos t)$, then $\left(\frac{d^2 y}{dx^2} \right)_{t=\frac{\pi}{4}} =$

$x = \sqrt{2} e^t (\sin t - \cos t)$ మరియు $y = \sqrt{2} e^t (\sin t + \cos t)$ అయితే, $\left(\frac{d^2 y}{dx^2} \right)_{t=\frac{\pi}{4}} =$

Options :

1. ✓ $-e^{-\pi/4}$

Maths - Question 64 (continued)

Original paper page 62; original question and option regions

$$\sqrt{2} e^{\pi/4}$$

$$\sqrt{2} e^{-\pi/4}$$

$$e^{-\pi/4}$$

Maths - Question 65

Original paper page 62; original question and option regions

P and Q are the ends of a diameter of the circle $x^2 + y^2 = a^2 \left(a > \frac{1}{\sqrt{2}} \right)$. s and t are the lengths of the perpendiculars drawn from P and Q onto the line $x + y = 1$ respectively. When the product st is maximum, the greater value among s, t is

వృత్తం $x^2 + y^2 = a^2 \left(a > \frac{1}{\sqrt{2}} \right)$ యొక్క వ్యాసం కొనలు P మరియు Q. s మరియు t లు వరుసగా

P, Q ల నుండి $x + y = 1$ రేఖ పైకి గీచిన లంబ దూరములు. st లబ్ధం గరిష్ఠం అయినప్పుడు s, t లలో పెద్ద దాని విలువ

Options :

$$a + \sqrt{2}$$

1. ✖

Maths - Question 65 (continued)

Original paper page 63; original question and option regions

$$a + \frac{1}{\sqrt{2}}$$

$$a - \frac{1}{\sqrt{2}}$$

$$a - \sqrt{2}$$

Maths - Question 66

Original paper page 63; original question and option regions

Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$ be such that $x = 0$ is the only real root of $P'(x) = 0$. If $P(-1) < P(1)$, then in the interval $[-1, 1]$

$P(x) = x^4 + ax^3 + bx^2 + cx + d$ అనేది $P'(x) = 0$ కు $x = 0$ మాత్రమే వాస్తవ మూలమయ్యేటట్లుగా ఉండనుకుందాం. $P(-1) < P(1)$ అయితే, $[-1, 1]$ అంతరంలో

Options :

$P(-1)$ is not minimum of $P(x)$, but $P(1)$ is the maximum of $P(x)$

$P(x)$ యొక్క కనిష్ఠం $P(-1)$ కాదు, అయితే, $P(x)$ యొక్క గరిష్ఠం $P(1)$

1. ✓

Maths - Question 66 (continued)

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$P(-1)$ is minimum of $P(x)$, but $P(1)$ is not the maximum of $P(x)$

$P(x)$ యొక్క కనిష్ఠం $P(-1)$, కాని $P(x)$ యొక్క గరిష్ఠం $P(1)$ కాదు

Neither $P(-1)$ is the minimum nor $P(1)$ is the maximum of $P(x)$

$P(x)$ కి, $P(-1)$ కనిష్ఠమూ కాదు, $P(1)$ గరిష్ఠమూ కాదు

$P(-1)$ is the minimum and $P(1)$ is the maximum of $P(x)$

$P(x)$ యొక్క కనిష్ఠం $P(-1)$ మరియు గరిష్ఠం $P(1)$

Maths - Question 68

Original paper page 65; original question and option regions

If the lengths of the tangent, subtangent, normal and subnormal for the curve $y = x^2 + x - 1$ at the point $(1, 1)$ are a, b, c and d respectively, then their increasing order is

బిందువు $(1, 1)$ వద్ద $y = x^2 + x - 1$ వక్రం యొక్క స్పర్శరేఖ, ఉప స్పర్శరేఖ, అభిలంబరేఖ మరియు ఉప అభిలంబరేఖల యొక్క పొడవులు వరుసగా a, b, c, d అయితే, వాటి ఆరోహణ క్రమం

Options :

b, d, a, c

1. ✖

2. ✖ b, a, c, d

3. ✖ a, b, c, d

4. ✔ b, a, d, c

Maths - Question 69

Original paper page 66; original question and option regions

$$\int \frac{x+1}{x^3-1} dx =$$

Options :

$$\frac{1}{3} \log \left(\frac{x+1}{x^2+x+1} \right) + c$$

1. ✖

$$\frac{1}{3} \log \left(\frac{(x-1)^2}{x^2+x+1} \right) + c$$

2. ✔

$$\frac{1}{3} \log \left(\frac{x-1}{x^2+x+1} \right) + c$$

3. ✖

Maths - Question 69 (continued)

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$$\frac{1}{3} \log \left(\frac{(x+1)^2}{x^2 - x + 1} \right) + c$$

Maths - Question 70

Original paper page 67; original question and option regions

$$\int \frac{x^4 - 16x^2 + 2x + 8}{x^3 - 4x^2 + 2} dx =$$

Options :

1.  $\frac{x^2 + 8x + c}{2}$

2.  $x^2 + 8x + c$

3.  $x^3 - 4x + c$

4.  $\frac{x^2 - 8x + c}{2}$

Maths - Question 71

Original paper page 68; original question and option regions

$$\int \frac{\sec^2 x}{(\sec x + \tan x)^{5/2}} dx =$$

Options :

1. ✖ $-\frac{(\sec x + \tan x)^{5/2}}{5} - \frac{(\sec x + \tan x)^{7/2}}{7} + c$

2. ✖ $-\frac{(\sec x - \tan x)^{5/2}}{5} - \frac{(\sec x - \tan x)^{7/2}}{7} + c$

3. ✖ $-\frac{(\sec x + \tan x)^{3/2}}{3} - \frac{(\sec x + \tan x)^{7/2}}{7} + c$

4. ✔ $-\frac{(\sec x - \tan x)^{3/2}}{3} - \frac{(\sec x - \tan x)^{7/2}}{7} + c$

Maths - Question 72

Original paper page 69; original question and option regions

$$\int \frac{1}{\cos x} \left[\frac{1}{\sin x} - \frac{1}{\sin x + 3 \cos x} \right] dx =$$

Options :

1. ✖ $\frac{1}{3} \log \left| \frac{\sin x}{\sin x + 3 \cos x} \right| + c$

2. ✖ $\log \left| \frac{\cos x}{\sin x + 3 \cos x} \right| + c$

3. ✖ $\frac{1}{3} \log \left| \frac{\cos x}{\sin x + 3 \cos x} \right| + c$

4. ✔ $\log \left| \frac{\sin x}{\sin x + 3 \cos x} \right| + c$

Maths - Question 73

Original paper page 69; original question and option regions

$$\int \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) dx =$$

Maths - Question 73 (continued)

Original paper page 70; original question and option regions

$$2[x \tan^{-1} x - \log \sqrt{1+x^2}] + c$$

$$2x \tan^{-1} x + \log \sqrt{1+x^2} + c$$

$$x \tan^{-1} x + \log \sqrt{1-x^2} + c$$

$$2[\tan^{-1} x - \log \sqrt{1+x^2}] + c$$

Maths - Question 74

Original paper page 70; original question and option regions

$$\int_0^x \frac{t^2}{\sqrt{a^2 + t^2}} dt =$$

Options :

1. ✖ $\frac{x}{2} \sqrt{a^2 + x^2} + \log |x + \sqrt{a^2 + x^2}|$

2. ✖ $\sqrt{a^2 + x^2} - a^2 \sinh^{-1} \frac{x}{a}$

Maths - Question 74 (continued)

Original paper page 71; original question and option regions

$$\frac{x}{2}\sqrt{a^2 + x^2} + \frac{a^2}{4}\log |x + \sqrt{a^2 + x^2}|$$

$$\frac{x}{2}\sqrt{a^2 + x^2} - \frac{a^2}{2}\sinh^{-1}\frac{x}{a}$$

Maths - Question 75

Original paper page 71; original question and option regions

The area (in sq. units) of the region bounded by the lines $x = 0$, $x = \frac{\pi}{2}$ and

$f(x) = \sin x$, $g(x) = \cos x$ is

రేఖలు $x = 0$, $x = \frac{\pi}{2}$ మరియు $f(x) = \sin x$, $g(x) = \cos x$ లచే పరిబద్ధమైన ప్రాంతం యొక్క

వైశాల్యం (చ॥ యూనిట్లలో)

Options :

1. ✓ $2(\sqrt{2} - 1)$

2. ✗ $2(\sqrt{3} - 1)$

3. ✗ $2(\sqrt{2} + 1)$

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Maths - Question 75 (continued)

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$$3\sqrt{2} + 1$$

Maths - Question 76

Original paper page 72; original question and option regions

$$\int_{\pi/6}^{\pi/3} \cos^{-4} x \, dx =$$

Options :

1. ✖ $\frac{64}{9\sqrt{3}}$

2. ✖ $\frac{52\sqrt{3}}{9}$

3. ✖ $\frac{62\sqrt{3}}{9}$

4. ✔ $\frac{44}{9\sqrt{3}}$

Maths - Question 77

Original paper page 73; original question and option regions

$$\int_0^{3\pi/2} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx =$$

Options :

1. ✖ 0

2. ✖ 1

3. ✖ $\frac{\pi}{4}$

4. ✔ $\frac{3\pi}{4}$

Maths - Question 79

Original paper page 74; original question and option regions

The differential equation for which $y^2 = 4a(x + a)$ (a is the parameter) is the general solution is

$y^2 = 4a(x + a)$ (a పరామితి) ని సాధారణ సాధన గా కలిగిన అవకలన సమీకరణం

Options :

$$y = 2x \frac{dy}{dx} + y \left(\frac{dy}{dx} \right)^2$$

1. ✓

$$y = y \frac{dy}{dx} - x \left(\frac{dy}{dx} \right)^2$$

2. ✗

Maths - Question 79 (continued)

Original paper page 75; original question and option regions

$$x = 3 \frac{dy}{dx} + y \left(\frac{dy}{dx} \right)^2$$

$$y = 3x^2 \frac{dy}{dx} + y^2 \left(\frac{dy}{dx} \right)^2$$

Maths - Question 80

Original paper page 75; original question and option regions

The general solution of the differential equation $\frac{dy}{dx} = \frac{2xy - 4x + y - 2}{2xy + x - 4y - 2}$ is

అవకలన సమీకరణం $\frac{dy}{dx} = \frac{2xy - 4x + y - 2}{2xy + x - 4y - 2}$ యొక్క సాధారణ సాధన

Options :

$$5(y - x) + 2 \log \left(\frac{y - 2}{x - 2} \right) = c$$

1. ✖

$$2(y - x) - 5 \log \left(\frac{y - 2}{x - 2} \right) = c$$

2. ✖

Maths - Question 80 (continued)

Original paper page 76; original question and option regions

$$2(y - x) + 5 \log \left(\frac{y - 2}{x - 2} \right) = c$$

$$5(y - x) - 2 \log \left(\frac{y - 2}{x - 2} \right) = c$$

Maths - Question 1

Original paper page 2; original question and option regions

$[t]$ denotes the greatest integer function and $[t - m] = [t] - m$ when $m \in \mathbb{Z}$.
If $k = 2[2x - 1] - 1$ and $3[2x - 2] + 1 = 2[2x - 1] - 1$ then the range of $f(x) = [k + 5x]$ is

$[t]$ గరిష్ఠ పూర్ణాంక ప్రమేయమును సూచిస్తుంది మరియు $m \in \mathbb{Z}$ అయినప్పుడు $[t - m] = [t] - m$.
 $k = 2[2x - 1] - 1$ మరియు $3[2x - 2] + 1 = 2[2x - 1] - 1$ అయితే, $f(x) = [k + 5x]$ యొక్క
వ్యాప్తి

Maths - Question 1 (continued)

Original paper page 3; original question and option regions

$$\{7, 8, 9\}$$

$$\{4, 5, 6\}$$

$$\{5, 6, 7\}$$

$$\{6, 7, 8\}$$

Maths - Question 2

Original paper page 3; original question and option regions

If $f(x) = (x + 1)^2 - 1$, $x \geq -1$, then $\{x \mid f(x) = f^{-1}(x)\}$ is

$f(x) = (x + 1)^2 - 1$, $x \geq -1$ అయితే, అప్పుడు $\{x \mid f(x) = f^{-1}(x)\}$ అనేది

Options :

1. ✓ $\{0, -1\}$

2. ✗ $\{-1, 0, 1\}$

Maths - Question 2 (continued)

Original paper page 4; original question and option regions

$$\left\{-1, 0, \frac{-3 + \sqrt{3}i}{2}, \frac{-3 - \sqrt{3}i}{2}\right\}$$

an empty set (ఒక సూన్య సమితి)

Maths - Question 3

Original paper page 4; original question and option regions

If $11^{12} - 11^2 = k(5 \times 10^9 + 6 \times 10^9 + 33 \times 10^8 + 110 \times 10^7 + \dots + 33)$, then $k =$

$11^{12} - 11^2 = k(5 \times 10^9 + 6 \times 10^9 + 33 \times 10^8 + 110 \times 10^7 + \dots + 33)$ అయితే, $k =$

Options :

1. ✖ 20

2. ✖ 50

3. ✖ 100

4. ✔ 200

Maths - Question 4

Original paper page 5; original question and option regions

If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a matrix A and $\det A = 4$, then the value of α is

ఒక మాత్రిక A యొక్క అనుబంధ మాత్రిక $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ మరియు $\det A = 4$ అయితే, α

యొక్క విలువ

Options :

1. ✖ 3

2. ✖ 22

3. ✔ 11

4. ✖ 4

Maths - Question 5

Original paper page 6; original question and option regions

If α is a real root of the equation $x^3 + 6x^2 + 5x - 42 = 0$, then the determinant of

the matrix $\begin{bmatrix} \alpha - 1 & \alpha + 1 & \alpha + 2 \\ \alpha - 2 & \alpha + 3 & \alpha - 3 \\ \alpha + 4 & \alpha - 4 & \alpha + 5 \end{bmatrix}$ is

α అనేది $x^3 + 6x^2 + 5x - 42 = 0$ సమీకరణం యొక్క వాస్తవ మూలమైతే,

$\begin{bmatrix} \alpha - 1 & \alpha + 1 & \alpha + 2 \\ \alpha - 2 & \alpha + 3 & \alpha - 3 \\ \alpha + 4 & \alpha - 4 & \alpha + 5 \end{bmatrix}$ మాత్రిక యొక్క నిర్ధారకము

Options :

1. ✖ 90
2. ✖ 120
3. ✔ -105
4. ✖ -135

Maths - Question 6

Original paper page 7; original question and option regions

The rank of the matrix $\begin{bmatrix} 2 & -3 & 4 & 0 \\ 5 & -4 & 2 & 1 \\ 1 & -3 & 5 & -4 \end{bmatrix}$ is

$\begin{bmatrix} 2 & -3 & 4 & 0 \\ 5 & -4 & 2 & 1 \\ 1 & -3 & 5 & -4 \end{bmatrix}$ మాత్రిక యొక్క ర్యాంక్

Options :

1. ✖ 0

2. ✔ 3

3. ✖ 2

4. ✖ 1

Maths - Question 7

Original paper page 8; original question and option regions

If z is a complex number such that $\frac{z-1}{z-i}$ is purely imaginary and locus of z

represents a circle with centre (α, β) and radius r ; then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} =$

$\frac{z-1}{z-i}$ శుద్ధ కల్పిత సంఖ్య అయ్యేటట్లు z ఒక సంకీర్ణ సంఖ్య మరియు z యొక్క బిందుపథం, (α, β)

కేంద్రము గానూ, r వ్యాసార్థం గానూ కలిగిన వృత్తమైతే, $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} =$

Options :

1. ✖ $4r$

2. ✖ r^2

3. ✖ $2r^2$

4. ✔ $4r^2$

Maths - Question 8

Original paper page 9; original question and option regions

If the least positive integer n satisfying the equation $\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^n = -1$ is p and the

least positive integer m satisfying the equation $\left(\frac{1-\sqrt{3}i}{1+\sqrt{3}i}\right)^m = \text{cis } \frac{2\pi}{3}$ is q , then

$$\sqrt{p^2 + q^2} =$$

p అనేది $\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^n = -1$ సమీకరణాన్ని తృప్తిపరిచే కనిష్ట ధనపూర్ణ సంఖ్య n మరియు q అనేది

$\left(\frac{1-\sqrt{3}i}{1+\sqrt{3}i}\right)^m = \text{cis } \frac{2\pi}{3}$ సమీకరణాన్ని తృప్తిపరిచే కనిష్ట ధన పూర్ణసంఖ్య m అయితే, $\sqrt{p^2 + q^2} =$

Options :

1. ✖ 5

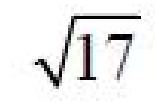
2. ✖ 10

3. ✔ $\sqrt{13}$

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Maths - Question 8 (continued)

Original paper page 10; original question and option regions


$$\sqrt{17}$$

Maths - Question 9

Original paper page 10; original question and option regions

Sum of the squares of the imaginary roots of the equation $z^8 - 20z^4 + 64 = 0$ is

$z^8 - 20z^4 + 64 = 0$ సమీకరణం యొక్క కల్పిత మూలాల వర్గాల మొత్తం

Options :

1. ✖ 0

2. ✔ -12

3. ✖ -4

4. ✖ -16

Maths - Question 10

Original paper page 11; original question and option regions

Let $(a - 3)x^2 + 12x + (a + 6) > 0, \forall x \in \mathbb{R} \text{ \& } a \in (\ell, \infty)$. If α is the least positive integral value of a , then the roots of $(\alpha - 3)x^2 + 12x + (\ell + 2) = 0$ are

అన్ని $x \in \mathbb{R}$ మరియు $a \in (\ell, \infty)$ లకు $(a - 3)x^2 + 12x + (a + 6) > 0$ అనుకోండి.

a యొక్క కనిష్ఠ ధనపూర్ణాంక విలువ α అయితే, $(\alpha - 3)x^2 + 12x + (\ell + 2) = 0$ యొక్క మూలాలు

Options :

1. ✖ 1, 2

2. ✖ 2, 3

3. ✔ -1, -2

4. ✖ -2, -3

Maths - Question 11

Original paper page 12; original question and option regions

If the roots of the equation $x^2 + 2ax + b = 0$ are real, distinct and differ at most by $2m$, then b lies in the interval

సమీకరణం $x^2 + 2ax + b = 0$ యొక్క మూలములు విభిన్న వాస్తవ సంఖ్యలై మరియు వాటి మధ్య గల భేదం $2m$ కు మించనిదైతే b ఉండే అంతరం

Options :

1. ✖ $(a^2, a^2 + m^2]$

2. ✖ $(a^2 + m^2, a^2)$

3. ✖ $[a^2, a^2 + 2m^2]$

4. ✔ $[a^2 - m^2, a^2)$

Maths - Question 12

Original paper page 12; original question and option regions

The cubic equation whose roots are the squares of the roots of the equation

$$x^3 - 2x^2 + 3x - 4 = 0 \text{ is}$$

$$x^3 - 2x^2 + 3x - 4 = 0 \text{ సమీకరణం యొక్క మూలాల వర్గాలను మూలాలుగా గలిగిన ఘన సమీకరణం}$$

Maths - Question 12 (continued)

Original paper page 13; original question and option regions

$$x^3 + 2x^2 + 7x - 16 = 0$$

$$x^3 + 2x^2 - 7x - 16 = 0$$

$$x^3 - 2x^2 - 7x + 16 = 0$$

$$x^3 - 2x^2 + 7x + 16 = 0$$

Maths - Question 13

Original paper page 13; original question and option regions

If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, then $\alpha^3 + \beta^3 + \gamma^3 =$

$x^3 + px^2 + qx + r = 0$ సమీకరణం యొక్క మూలాలు α, β, γ అయితే, $\alpha^3 + \beta^3 + \gamma^3 =$

Options :

1. ✖ $p^3 - 3pq + r$

2. ✖ $p^2 - 2pq + r$

Maths - Question 13 (continued)

Original paper page 14; original question and option regions

$$3pq - 3r - p^3$$

$$3pq + 3r + p^3$$

Maths - Question 17

Original paper page 16; original question and option regions

If $C_0, C_1, C_2, \dots, C_n$ are the binomial coefficients in the expansion of $(1 + x)^n$, then $(C_0 + C_1) - (C_2 + C_3) + (C_4 + C_5) - (C_6 + C_7) + \dots =$

$(1 + x)^n$ విస్తరణలో $C_0, C_1, C_2, \dots, C_n$ లు ద్వీవద గుణకాలయితే,

$$(C_0 + C_1) - (C_2 + C_3) + (C_4 + C_5) - (C_6 + C_7) + \dots =$$

Maths - Question 17 (continued)

Original paper page 17; original question and option regions

$$2^{n/2} \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right)$$

$$2^{n/2} \left(\cos \frac{n\pi}{3} + \sin \frac{n\pi}{3} \right)$$

$$2^{n/2} \left(\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} \right)$$

$$2^{n/2} \left(\cos \frac{n\pi}{4} + \sin \frac{n\pi}{4} \right)$$

Maths - Question 18

Original paper page 17; original question and option regions

$$1 + \frac{4}{15} + \frac{4.10}{15.30} + \frac{4.10.16}{15.30.45} + \dots \quad \infty =$$

Maths - Question 18 (continued)

Original paper page 18; original question and option regions

$$\left(\frac{3}{5}\right)^{2/3}$$

$$\left(\frac{5}{3}\right)^{2/3}$$

$$\left(\frac{3}{5}\right)^{3/2}$$

$$\left(\frac{5}{3}\right)^{3/2}$$

Maths - Question 19

Original paper page 18; original question and option regions

If $\frac{3x+1}{(x-1)(x^2+2)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+2}$, then $5(A-B) =$

$\frac{3x+1}{(x-1)(x^2+2)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+2}$, అయితే, $5(A-B) =$

Maths - Question 19 (continued)

Original paper page 19; original question and option regions

$$A + C$$

$$8C$$

$$C + 8$$

$$\frac{C}{8}$$

Maths - Question 20

Original paper page 19; original question and option regions

$$\operatorname{cosec} 48^{\circ} + \operatorname{cosec} 96^{\circ} + \operatorname{cosec} 192^{\circ} + \operatorname{cosec} 384^{\circ} =$$

Options :

1. ✖ $4\sqrt{3}$

2. ✖ $-4\sqrt{3}$

3. ✔ 0

4. ✖ 1

Maths - Question 21

Original paper page 20; original question and option regions

If $\sqrt{3} \cos \theta + \sin \theta > 0$ then

$\sqrt{3} \cos \theta + \sin \theta > 0$ అయితే, అప్పుడు

Options :

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

1. ✖

$$-\frac{\pi}{3} < \theta < \frac{2\pi}{3}$$

2. ✔

$$-\frac{2\pi}{3} < \theta < \frac{\pi}{3}$$

3. ✖

$$-\frac{\pi}{6} < \theta < \frac{5\pi}{6}$$

4. ✖

Maths - Question 22

Original paper page 21; original question and option regions

If $\cos \theta = \frac{-3}{5}$ and θ does not lie in second quadrant, then $\tan \frac{\theta}{2} =$

$\cos \theta = \frac{-3}{5}$ మరియు θ రెండవ పాదంలో లేని కోణము అయితే, $\tan \frac{\theta}{2} =$

Options :

1. ✖ 2

2. ✖ 1

3. ✔ -2

4. ✖ -1

Maths - Question 23

Original paper page 21; original question and option regions

The general solution satisfying both the equations $\sin x = -\frac{3}{5}$ and $\cos x = -\frac{4}{5}$ is

$\sin x = -\frac{3}{5}$ మరియు $\cos x = -\frac{4}{5}$ సమీకరణాలు రెండింటినీ తృప్తిపరిచే సాధారణ సాధన

Maths - Question 23 (continued)

Original paper page 22; original question and option regions

$$x = (2n + 1)\pi + \tan^{-1}\left(\frac{3}{4}\right), n \in \mathbb{Z}$$

$$x = 2n\pi + \tan^{-1}\left(\frac{3}{4}\right), n \in \mathbb{Z}$$

$$x = n\pi + \tan^{-1}\left(\frac{3}{4}\right), n \in \mathbb{Z}$$

$$x = n\pi \pm \tan^{-1}\left(\frac{3}{4}\right), n \in \mathbb{Z}$$

Maths - Question 24

Original paper page 22; original question and option regions

The number of solutions of $\tan^{-1} 1 + \frac{1}{2} \cos^{-1} x^2 - \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right) = 0$

is

$$\tan^{-1} 1 + \frac{1}{2} \cos^{-1} x^2 - \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right) = 0 \quad \text{యొక్క సాధనల సంఖ్య}$$

Maths - Question 24 (continued)

Original paper page 23; original question and option regions

3

0

1

infinitely many (అనంతమైనన్ని)

Maths - Question 25

Original paper page 23; original question and option regions

$$\tanh^{-1}(\sin \theta) =$$

Options :

1. ✖ $\sinh^{-1}(\operatorname{cosec} \theta)$

2. ✖ $\sinh^{-1}(\sec \theta)$

3. ✖ $\cosh^{-1}(\operatorname{cosec} \theta)$

4. ✔ $\cosh^{-1}(\sec \theta)$

Maths - Question 26

Original paper page 24; original question and option regions

In $\triangle ABC$, if $a = 8$, $b = 10$, $c = 12$, then $\frac{r}{R} =$

$\triangle ABC$ లో $a = 8$, $b = 10$, $c = 12$ అయితే, $\frac{r}{R} =$

Options :

1. ✖ $\frac{8}{15}$

2. ✔ $\frac{7}{16}$

3. ✖ $\frac{3}{5}$

4. ✖ $\frac{5}{8}$

Maths - Question 27

Original paper page 25; original question and option regions

In triangle ABC, if $a = 13$, $b = 8$, $c = 7$, then $\cos (B + C) =$

త్రిభుజం ABC లో if $a = 13$, $b = 8$, $c = 7$ అయితే, $\cos (B + C) =$

Options :

1. ✖ $\frac{11}{13}$

2. ✖ $\frac{23}{26}$

3. ✖ $\frac{3}{4}$

4. ✔ $\frac{1}{2}$

Maths - Question 28

Original paper page 25; original question and option regions

In a triangle ABC, if $(r_1 - r_3)(r_1 - r_2) - 2r_2r_3 = 0$, then $a^2 - b^2 =$

ఒక త్రిభుజం ABC లో $(r_1 - r_3)(r_1 - r_2) - 2r_2r_3 = 0$ అయితే, $a^2 - b^2 =$

Maths - Question 28 (continued)

Original paper page 26; original question and option regions

$$c^2 + \frac{b^2}{4}$$

$$c^2$$

$$abc$$

$$\frac{b+a}{c}$$

Maths - Question 30

Original paper page 27; original question and option regions

If $\vec{a} = 2\vec{i} - 3\vec{j} + 5\vec{k}$ and $\vec{b} = -\vec{i} + 3\vec{j} + 3\vec{k}$ are two vectors, then the vector of magnitude 28 units in the direction of the vector $\vec{a} - \vec{b}$ is

$\vec{a} = 2\vec{i} - 3\vec{j} + 5\vec{k}$ మరియు $\vec{b} = -\vec{i} + 3\vec{j} + 3\vec{k}$ లు రెండు సదిశలైతే, 28 యూనిట్ల పరిమాణం కలిగి $\vec{a} - \vec{b}$ సదిశ దిశలో ఉన్న సదిశ

Options :

1. ✖ $3\vec{i} + 6\vec{j} - 2\vec{k}$

2. ✔ $12\vec{i} - 24\vec{j} + 8\vec{k}$

3. ✖ $3\vec{i} - 6\vec{j} - 2\vec{k}$

Maths - Question 30 (continued)

Original paper page 28; original question and option regions

$$12\bar{i} + 24\bar{j} - 8\bar{k}$$

Maths - Question 31

Original paper page 28; original question and option regions

If $\bar{a}$ is a unit vector, then $|\bar{a} \times \bar{i}|^2 + |\bar{a} \times \bar{j}|^2 + |\bar{a} \times \bar{k}|^2 =$

$\bar{a}$ ఒక యూనిట్ సదిశ అయితే, $|\bar{a} \times \bar{i}|^2 + |\bar{a} \times \bar{j}|^2 + |\bar{a} \times \bar{k}|^2 =$

Options :

1. ✖ 4

2. ✖ 1

3. ✖ 0

4. ✔ 2

Maths - Question 32

Original paper page 29; original question and option regions

If $\vec{a} = \vec{i} - 2\vec{j} - 3\vec{k}$, $\vec{b} = -2\vec{i} + 3\vec{j} + 4\vec{k}$, $\vec{c} = 5\vec{i} - 4\vec{j} + 3\vec{k}$ and $\vec{d} = 3\vec{i} + \vec{j} + 5\vec{k}$ are four vectors then $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) =$

$\vec{a} = \vec{i} - 2\vec{j} - 3\vec{k}$, $\vec{b} = -2\vec{i} + 3\vec{j} + 4\vec{k}$, $\vec{c} = 5\vec{i} - 4\vec{j} + 3\vec{k}$ మరియు $\vec{d} = 3\vec{i} + \vec{j} + 5\vec{k}$ లు నాలుగు సదిశలైతే $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) =$

Options :

1. ✓ $18\vec{i} + 6\vec{j} + 30\vec{k}$

2. ✗ $8\vec{i} - 3\vec{j} + 8\vec{k}$

3. ✗ $19\vec{i} - 5\vec{j} + 21\vec{k}$

4. ✗ $27\vec{i} - 8\vec{j} + 29\vec{k}$

Maths - Question 33

Original paper page 30; original question and option regions

$3\bar{i} + \bar{j} + \bar{k}$, $2\bar{i} + \bar{k}$, $\bar{i} + 5\bar{j}$ are the position vectors of three non collinear points A, B, C respectively. If the perpendicular drawn from C onto $\overline{AB}$ meets $\overline{AB}$ at the point $a\bar{i} + b\bar{j} + c\bar{k}$, then $a + b + c =$

$3\bar{i} + \bar{j} + \bar{k}$, $2\bar{i} + \bar{k}$, $\bar{i} + 5\bar{j}$ లు వరుసగా సరేఖీయాలు కాని మూడు బిందువులు A, B, C ల స్థాన సదిశలు. సదిశ $\overline{AB}$ పైకి C నుండి గీసిన లంబం $\overline{AB}$ ను $a\bar{i} + b\bar{j} + c\bar{k}$ అనే బిందువు వద్ద ఖండిస్తే, $a + b + c =$

Options :

1. ✖ 5
2. ✖ 3
3. ✔ 7
4. ✖ 9

Maths - Question 34

Original paper page 31; original question and option regions

Let $x_1, x_2, \dots, x_{11}$ be the observations satisfying $\sum_{i=1}^{11} (x_i - 4) = 22$ and

$\sum_{i=1}^{11} (x_i - 4)^2 = 154$. If the mean and variance of the observations are α and β ,

then the quadratic equation having the roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is

$x_1, x_2, \dots, x_{11}$ అనే పరిశీలనలు $\sum_{i=1}^{11} (x_i - 4) = 22$ మరియు $\sum_{i=1}^{11} (x_i - 4)^2 = 154$ లను తృప్తి పరుస్తున్నాయనుకోండి. ఆ పరిశీలనల మధ్యమం మరియు విస్తృతులు వరుసగా α మరియు β లయితే, $\frac{\alpha}{\beta}$ మరియు $\frac{\beta}{\alpha}$ లు మూలాలుగా గల వర్గసమీకరణం.

Options :

$$15x^2 - 16x + 15 = 0$$

1. ✖

$$15x^2 - 34x + 15 = 0$$

2. ✔

$$x^2 - 16x + 60 = 0$$

3. ✖

$$12x^2 - 25x + 20 = 0$$

4. ✖

Maths - Question 38

Original paper page 35; original question and option regions

A bag contains 5 balls of unknown colours. There are equal chances that out of these five balls, there may be 0 or 1 or 2 or 3 or 4 or 5 red balls. A ball is taken out from the bag at random and is found to be red. The probability that it is the only red ball in the bag is

ఒక సంచిలో రంగు తెలియని బంతులు 5 ఉన్నాయి. ఈ 5 బంతులలో 0 లేదా 1 లేదా 2 లేదా 3 లేదా 4 లేదా 5 బంతులు ఎర్రనివి కావడానికి సమానమైన సంభావ్యతలున్నాయి. ఆ సంచి నుంచి ఒక బంతిని యాదృచ్ఛికంగా బయటకు తీసి దానిని ఎర్రని బంతిగా గుర్తించారు. ఆ సంచిలో అది ఒక్కటే ఎర్రని బంతి కావడానికి గల సంభావ్యత

Options :

1. ✖ $\frac{1}{5}$

2. ✖ $\frac{1}{6}$

3. ✔ $\frac{1}{15}$

4. ✖ $\frac{1}{30}$

Maths - Question 42

Original paper page 39; original question and option regions

After the coordinate axes are rotated through an angle $\frac{\pi}{4}$ in the anti clockwise direction without shifting the origin, if the equation $x^2 + y^2 - 2x - 4y - 20 = 0$ transforms to $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ in the new coordinate system,

then $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} =$

మూల బిందువును మార్చకుండా, నిరూపక అక్షాలను ధనదిశలో $\frac{\pi}{4}$ కోణంతో భ్రమణం చేసినప్పుడు సమీకరణం $x^2 + y^2 - 2x - 4y - 20 = 0$, నూతన నిరూపక వ్యవస్థలో

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ గా రూపాంతరం చెందితే $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} =$

Options :

1. ✖ -20

2. ✔ -25

3. ✖ -30

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Maths - Question 42 (continued)

Original paper page 40; original question and option regions

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Maths - Question 43

Original paper page 40; original question and option regions

$A(-2, 3)$ is a point on the line $4x + 3y - 1 = 0$. If the points on the line that are 10 units away from the point A are (x_1, y_1) and (x_2, y_2) , then $(x_1 + y_1)^2 + (x_2 + y_2)^2 =$

$4x + 3y - 1 = 0$ రేఖపై $A(-2, 3)$ ఒక బిందువు. ఆ రేఖపై A బిందువు నుండి 10 యూనిట్ల దూరంలో గల బిందువులు (x_1, y_1) మరియు (x_2, y_2) అయితే, $(x_1 + y_1)^2 + (x_2 + y_2)^2 =$

Options :

1. ✓ 10
2. ✗ 90
3. ✗ 180
4. ✗ 405

Maths - Question 45

Original paper page 42; original question and option regions

If the equation of the pair of lines passing through $(1, 1)$ and perpendicular to the pair of lines $2x^2 + xy - y^2 - x + 2y - 1 = 0$ is

$$ax^2 + 2hxy + by^2 + 2gx + 3y = 0, \text{ then } \frac{b}{a} =$$

$(1, 1)$ బిందువు గుండా పోతూ $2x^2 + xy - y^2 - x + 2y - 1 = 0$ రేఖాయుగ్మానికి లంబంగా ఉండే

రేఖాయుగ్మం యొక్క సమీకరణం $ax^2 + 2hxy + by^2 + 2gx + 3y = 0$ అయితే, $\frac{b}{a} =$

Options :

1. ✖ g/h
2. ✔ $2(g+h)$
3. ✖ $2(g-h)$
4. ✖ gh

Maths - Question 46

Original paper page 43; original question and option regions

If the combined equation of the lines joining the origin to the points of intersection of the curve $x^2 + y^2 - 2x - 4y + 2 = 0$ and the line $x + y - 2 = 0$ is $(l_1x + m_1y)(l_2x + m_2y) = 0$, then $l_1 + l_2 + m_1 + m_2 =$

వక్రం $x^2 + y^2 - 2x - 4y + 2 = 0$ మరియు రేఖ $x + y - 2 = 0$ ల ఖండన బిందువులను మూలబిందువుతో కలుపగా ఏర్పడిన రేఖల ఉమ్మడి సమీకరణము $(l_1x + m_1y)(l_2x + m_2y) = 0$ అయితే, $l_1 + l_2 + m_1 + m_2 =$

Options :

1. ✖ 16
2. ✖ -6
3. ✔ -2
4. ✖ 10

Maths - Question 47

Original paper page 44; original question and option regions

If the circles $x^2 + y^2 + 5kx + 2y + k = 0$ and $2x^2 + 2y^2 + 2kx + 3y - 1 = 0$, $k \in \mathbb{R}$ intersect at points P and Q, then the line $4x + 5y - k = 0$ passes through P and Q for

$x^2 + y^2 + 5kx + 2y + k = 0$ మరియు $2x^2 + 2y^2 + 2kx + 3y - 1 = 0$, $k \in \mathbb{R}$ వృత్తములు P మరియు Q బిందువుల వద్ద ఖండించుకొంటే, అప్పుడు $4x + 5y - k = 0$ రేఖ

Options :

exactly one value of k

1. ✖ ఖచ్చితముగా k యొక్క ఒకే ఒక విలువకు P మరియు Q గుండా పోతుంది.

exactly two values of k

2. ✖ ఖచ్చితముగా k యొక్క రెండు విలువలకు P మరియు Q గుండా పోతుంది.

no value of k

3. ✔ k యొక్క ఏ విలువకూ P మరియు Q గుండా పోదు.

infinitely many values of k

4. ✖ k యొక్క అపరిమితమైనట్టి విలువలకు P మరియు Q గుండా పోతుంది.

Maths - Question 48

Original paper page 45; original question and option regions

The slope of one of the direct common tangents drawn to the circles $x^2 + y^2 - 2x + 4y + 1 = 0$ and $x^2 + y^2 - 4x - 2y + 4 = 0$ is

$x^2 + y^2 - 2x + 4y + 1 = 0$ మరియు $x^2 + y^2 - 4x - 2y + 4 = 0$ వృత్తాలకు గీచిన ప్రత్యక్ష ఉమ్మడి స్పర్శ రేఖలలో ఒకదాని యొక్క వాలు

Options :

1. ✖ 0

2. ✔ $\frac{4}{3}$

3. ✖ $\frac{3}{4}$

4. ✖ 1

Maths - Question 49

Original paper page 46; original question and option regions

If $(1, a), (b, 2)$ are conjugate points with respect to the circle $x^2 + y^2 = 25$, then

$$4a + 2b =$$

$x^2 + y^2 = 25$ వృత్తం దృష్ట్యా $(1, a), (b, 2)$ బిందువులు సంయుగ్మ బిందువులు అయితే, $4a + 2b =$

Options :

1. ✖ 25

2. ✔ 50

3. ✖ 100

4. ✖ 150

Maths - Question 50

Original paper page 46; original question and option regions

If the pole of the line $x + 2by - 5 = 0$ with respect to the circle $S \equiv x^2 + y^2 - 4x - 6y + 4 = 0$ lies on the line $x + by + 1 = 0$, then the polar of the point $(b, -b)$ with respect to the circle $S = 0$ is

$S \equiv x^2 + y^2 - 4x - 6y + 4 = 0$ వృత్తము దృష్ట్యా $x + 2by - 5 = 0$ అనే సరళరేఖ యొక్క ధ్రువం $x + by + 1 = 0$ అనే సరళరేఖపై ఉంటే, $S = 0$ వృత్తం దృష్ట్యా $(b, -b)$ బిందువు యొక్క ధ్రువ రేఖ సమీకరణం

Maths - Question 50 (continued)

Original paper page 47; original question and option regions

$$5y - 6 = 0$$

$$y - 6 = 0$$

$$x + 5y - 6 = 0$$

$$5x + y - 6 = 0$$

Maths - Question 51

Original paper page 47; original question and option regions

If $P(\alpha, \beta)$ is the radical centre of the circles $S \equiv x^2 + y^2 + 4x + 7 = 0$, $S' \equiv 2x^2 + 2y^2 + 3x + 5y + 9 = 0$ and $S'' \equiv x^2 + y^2 + y = 0$, then the length of the tangent drawn from P to $S' = 0$ is

$S \equiv x^2 + y^2 + 4x + 7 = 0$, $S' \equiv 2x^2 + 2y^2 + 3x + 5y + 9 = 0$ మరియు

$S'' \equiv x^2 + y^2 + y = 0$ అనే వృత్తాల మూలకేంద్రం $P(\alpha, \beta)$ అయితే, P నుండి $S' = 0$ కు గీచిన స్పర్శరేఖ పొడవు

Options :

1. ✖ 5

Maths - Question 51 (continued)

Original paper page 48; original question and option regions

8

4

2

Maths - Question 52

Original paper page 48; original question and option regions

If the tangents of the parabola $y^2 = 8x$ passing through the point $P(1, 3)$ touches the parabola at A and B, then the area (in sq. units) of $\triangle ABC$ is

$y^2 = 8x$ వరాచలయానికి $P(1, 3)$ బిందువు గుండా పోవునట్లు గీసిన స్పర్శరేఖలు, ఆ వరాచలయాన్ని A మరియు B ల వద్ద స్పృశిస్తే, $\triangle ABC$ వైశాల్యము (చ.యూనిట్లలో)

Options :

1. ✖ 1

2. ✖ $\frac{3}{4}$

Maths - Question 52 (continued)

Original paper page 49; original question and option regions

$$\frac{1}{2}$$

$$\frac{1}{4}$$

Maths - Question 53

Original paper page 49; original question and option regions

The equation of the normal drawn at the point $(\sqrt{2} + 1, -1)$ to the ellipse

$$x^2 + 2y^2 - 2x + 8y + 5 = 0 \text{ is}$$

$x^2 + 2y^2 - 2x + 8y + 5 = 0$ దీర్ఘ వృత్తానికి, $(\sqrt{2} + 1, -1)$ బిందువు వద్ద గీచిన అభిలంబరేఖ సమీకరణం

Options :

1. ✖ $x + y = \sqrt{2}$

2. ✖ $x - 2y = 3 + \sqrt{2}$

3. ✔ $\sqrt{2}x - y = 3 + \sqrt{2}$

Maths - Question 53 (continued)

Original paper page 50; original question and option regions

$$2x + y = 2\sqrt{2} + 1$$

Maths - Question 54

Original paper page 50; original question and option regions

If $3x + 2\sqrt{2}y + k = 0$ is a normal to the hyperbola $4x^2 - 9y^2 - 36 = 0$ making positive intercepts on both the axes, then $k =$

$4x^2 - 9y^2 - 36 = 0$ అతి పరావలయానికి, రెండు నిరూపకాక్షాలపైనా ధనాత్మక అంతర ఖండాలను చేసే ఒక అభిలంబ రేఖ $3x + 2\sqrt{2}y + k = 0$ అయితే, $k =$

Options :

1. ✖ $13\sqrt{2}$

2. ✖ $-5\sqrt{2}$

3. ✖ $-2\sqrt{2}$

4. ✔ $-13\sqrt{2}$

Maths - Question 57

Original paper page 52; original question and option regions

If the direction cosines of two lines satisfy the equations $l - 2m + n = 0$, $l/m + 10mn - 2n/l = 0$ and θ is the angle between the lines, then $\cos \theta =$

రెండు రేఖల దిక్ కోసైన్లు $l - 2m + n = 0$, $l/m + 10mn - 2n/l = 0$ సమీకరణాలను తృప్తిపరుస్తాయి మరియు ఆ రేఖల మధ్య కోణం θ అయితే, $\cos \theta =$

Maths - Question 57 (continued)

Original paper page 53; original question and option regions

$$\frac{\pi}{6}$$

$$\frac{8}{\sqrt{70}}$$

$$\frac{\pi}{3}$$

$$\frac{20}{3\sqrt{70}}$$

Maths - Question 59

Original paper page 54; original question and option regions

$$\lim_{x \rightarrow 0} \frac{x^2 \sin^2(3x) + \sin^4(6x)}{(1 - \cos 3x)^2} =$$

Options :

1.  $\frac{580}{9}$

2.  $\frac{145}{3}$

3.  $\frac{580}{3}$

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Maths - Question 59 (continued)

Original paper page 55; original question and option regions

145

9

Maths - Question 60

Original paper page 55; original question and option regions

If a real valued function $f(x) = \begin{cases} (1 + \sin x)^{\operatorname{cosec} x}, & -\pi/2 < x < 0 \\ a, & x = 0 \\ \frac{e^{2/x} + e^{3/x}}{ae^{2/x} + be^{3/x}}, & 0 < x < \pi/2 \end{cases}$,

is continuous at $x = 0$, then $ab =$

ఒక వాస్తవ మూల్య ప్రమేయము $f(x) = \begin{cases} (1 + \sin x)^{\operatorname{cosec} x}, & -\pi/2 < x < 0 \\ a, & x = 0 \\ \frac{e^{2/x} + e^{3/x}}{ae^{2/x} + be^{3/x}}, & 0 < x < \pi/2 \end{cases}$

$x = 0$ వద్ద అవిచ్ఛిన్నము అయితే, $ab =$

Options :

e

1. ✖

Maths - Question 60 (continued)

Original paper page 56; original question and option regions

$$e^2$$

$$1$$

$$-1$$

Maths - Question 61

Original paper page 56; original question and option regions

$$\lim_{x \rightarrow 0} \frac{(\operatorname{cosec} x - \cot x)(e^x - e^{-x})}{\sqrt{3} - \sqrt{2 + \cos x}} =$$

Options :

1. ✖ $3\sqrt{2}$

2. ✖ $2\sqrt{3}$

3. ✖ $3\sqrt{3}$

Maths - Question 61 (continued)

Original paper page 57; original question and option regions

$$4\sqrt{3}$$

Maths - Question 62

Original paper page 57; original question and option regions

If $y = \sqrt{\cosh x} + \sqrt{\cosh x}$, then $\frac{dy}{dx} =$

$$y = \sqrt{\cosh x} + \sqrt{\cosh x}, \quad \frac{dy}{dx} =$$

Options :

$$\frac{\sinh x (2y^2 + 2 \cosh x + 1)}{4y (y^2 + \cosh x)}$$

1. ✖

$$\frac{\sinh x (2y^2 - 2 \cosh x - 1)}{4y (y^2 - \cosh x)}$$

2. ✖

$$\frac{\sinh x (1 - 2\sqrt{\cosh x})}{4y\sqrt{\cosh x}}$$

3. ✖

Maths - Question 62 (continued)

Original paper page 58; original question and option regions

$$\frac{\sinh x (1 + 2\sqrt{\cosh x})}{4y\sqrt{\cosh x}}$$

Maths - Question 63

Original paper page 58; original question and option regions

If $y = \tan^{-1}\sqrt{x^2 - 1} + \sinh^{-1}\sqrt{x^2 - 1}$, $x > 1$, then $\frac{dy}{dx} =$

$y = \tan^{-1}\sqrt{x^2 - 1} + \sinh^{-1}\sqrt{x^2 - 1}$, $x > 1$ అయితే, $\frac{dy}{dx} =$

Options :

$$\frac{1}{x\sqrt{x^2 - 1}}$$

1. ✖

$$\frac{x + 1}{x\sqrt{x^2 - 1}}$$

2. ✔

$$\frac{x + 1}{x^2\sqrt{x^2 - 1}}$$

3. ✖

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Maths - Question 63 (continued)

Original paper page 59; original question and option regions

$$\frac{x}{\sqrt{x^2 - 1}}$$

Maths - Question 64

Original paper page 59; original question and option regions

If $y = (\log x)^{1/x} + x^{\log x}$, at $x = e$, $\frac{dy}{dx} =$

$y = (\log x)^{1/x} + x^{\log x}$ అయితే, $x = e$ వద్ద $\left(\frac{dy}{dx}\right) =$

Options :

$$2 + \frac{1}{e}$$

1. ✖

$$e^2 + \frac{1}{2}$$

2. ✖

$$\frac{1}{e^2} + 2$$

3. ✔

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Maths - Question 64 (continued)

Original paper page 60; original question and option regions

$$e + \frac{1}{e}$$

Maths - Question 65

Original paper page 60; original question and option regions

The interval in which the function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function, is

$f(x) = \tan^{-1}(\sin x + \cos x)$ ప్రమేయం ఆరోహణ ప్రమేయం అయ్యే అంతరం

Options :

$$\left(0, \frac{\pi}{2}\right)$$

1. ✖

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

2. ✖

$$\left(-\frac{3\pi}{4}, \frac{\pi}{4}\right)$$

3. ✔

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Maths - Question 65 (continued)

Original paper page 61; original question and option regions

$$\left(\frac{\pi}{4}, \frac{\pi}{2} \right)$$

Maths - Question 66

Original paper page 61; original question and option regions

The slope of a tangent drawn at the point $P(\alpha, \beta)$ lying on the curve $y = \frac{1}{2x-5}$ is

-2. If P lies in the fourth quadrant, then $\alpha - \beta =$

$y = \frac{1}{2x-5}$ వక్రంపై ఉన్న బిందువు $P(\alpha, \beta)$ వద్ద గీచిన స్పర్శరేఖ వాలు -2. P నాల్గవ పాదంలో ఉన్నట్లైతే, అప్పుడు $\alpha - \beta =$

Options :

1. ✖ 4

2. ✔ 3

3. ✖ 2

4. ✖ 1

Maths - Question 67

Original paper page 62; original question and option regions

The function $f(x) = xe^{-x} \forall x \in \mathbb{R}$ attains a maximum value at $x = k$, then $k =$

ప్రమేయము $f(x) = xe^{-x} \forall x \in \mathbb{R}$ ఒక గరిష్ట విలువను $x = k$ వద్ద పొందితే, $k =$

Options :

1. ✓ 1

2. ✗ 2

3. ✗ $\frac{1}{e}$

4. ✗ 3

Maths - Question 68

Original paper page 62; original question and option regions

If m and M are the absolute minimum and absolute maximum values of the function $f(x) = 2\sqrt{2} \sin x - \tan x$ in the interval $[0, \pi/3]$, then $m + M =$

m మరియు M లు $[0, \pi/3]$ అంతరంలో $f(x) = 2\sqrt{2} \sin x - \tan x$ ప్రమేయం యొక్క పరమ కనిష్ఠ విలువ మరియు పరమ గరిష్ఠ విలువలు అయితే, $m + M =$

Maths - Question 68 (continued)

Original paper page 63; original question and option regions

-1

0

1

2

Maths - Question 69

Original paper page 63; original question and option regions

$$\int \frac{\sec^2 x}{\sin^7 x} dx - \int \frac{7}{\sin^7 x} dx =$$

Options :

$$\frac{1}{\sin^6 x \cos x} + c$$

1. ✓

$$\frac{\tan x}{\sin^8 x} + c$$

2. ✗

$$\sin^8 x \cos x + c$$

3. ✗

Maths - Question 69 (continued)

Original paper page 64; original question and option regions

$$\sec x \tan^7 x + c$$

Maths - Question 70

Original paper page 64; original question and option regions

If $\int (x^6 + x^4 + x^2)\sqrt{2x^4 + 3x^2 + 6} \, dx = f(x) + c$, then $f(3) =$

$\int (x^6 + x^4 + x^2)\sqrt{2x^4 + 3x^2 + 6} \, dx = f(x) + c$ అయితే, $f(3) =$

Options :

1. ✖ $\frac{3}{2}(95)^{3/2}$

2. ✔ $\frac{3}{2}(195)^{3/2}$

3. ✖ $\frac{3}{2}(265)^{3/2}$

4. ✖ $\frac{3}{2}(175)^{3/2}$

Maths - Question 71

Original paper page 65; original question and option regions

$$\int \frac{dx}{(x+1)\sqrt{x^2+1}} =$$

Options :

1. ✖ $\frac{1}{\sqrt{2}} \sinh^{-1} \left(\frac{1+x}{1-x} \right) + c$

2. ✖ $\frac{1}{\sqrt{2}} \sinh^{-1} \left(\frac{1-x}{1+x} \right) + c$

3. ✔ $-\frac{1}{\sqrt{2}} \sinh^{-1} \left(\frac{1-x}{1+x} \right) + c$

4. ✖ $-\frac{1}{\sqrt{2}} \sinh^{-1} \left(\frac{1+x}{1-x} \right) + c$

Maths - Question 72

Original paper page 66; original question and option regions

If $\int \frac{dx}{2\cos x + 3\sin x + 4} = \frac{2}{\sqrt{3}}f(x) + c$, then $f\left(\frac{2\pi}{3}\right) =$

$\int \frac{dx}{2\cos x + 3\sin x + 4} = \frac{2}{\sqrt{3}}f(x) + c$ అయితే, $f\left(\frac{2\pi}{3}\right) =$

Options :

1. ✖ $\frac{\pi}{12}$

2. ✖ $\frac{\pi}{8}$

3. ✔ $\frac{5\pi}{12}$

4. ✖ $\frac{5\pi}{8}$

Maths - Question 73

Original paper page 67; original question and option regions

If $\int \frac{1}{\left((x+4)^3(x+1)^5\right)^{1/4}} dx = A \cdot \left(\frac{x+4}{x+1}\right)^n + c$, then

$\int \frac{1}{\left((x+4)^3(x+1)^5\right)^{1/4}} dx = A \cdot \left(\frac{x+4}{x+1}\right)^n + c$ అయితే, అప్పుడు

Options :

1. ✖ $n.A = 3$

2. ✔ $n + \frac{1}{A} = -\frac{1}{2}$

3. ✖ $A + n = 1$

4. ✖ $A = n$

Maths - Question 74

Original paper page 67; original question and option regions

$$\int_{-\pi/2}^{\pi/2} \sin^2 x \cos^2 x (\sin x + \cos x) dx =$$

Maths - Question 74 (continued)

Original paper page 68; original question and option regions

$$0$$

$$\frac{2}{15}$$

$$\frac{4}{15}$$

$$\frac{2}{5}$$

Maths - Question 75

Original paper page 68; original question and option regions

$$\int_{1/5}^{1/2} \frac{\sqrt{x - x^2}}{x^3} dx =$$

Options :

1. ✖ $\frac{21}{2}$

Maths - Question 75 (continued)

Original paper page 69; original question and option regions

$$\frac{14}{3}$$

$$\frac{7}{3}$$

$$\frac{7}{2}$$

Maths - Question 76

Original paper page 69; original question and option regions

$$\int_0^{400\pi} \sqrt{1 - \cos 2x} \, dx =$$

Options :

1. ✖ $100\sqrt{2}$

2. ✖ $200\sqrt{2}$

3. ✖ $400\sqrt{2}$

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Maths - Question 76 (continued)

Original paper page 70; original question and option regions

$$800\sqrt{2}$$

Maths - Question 77

Original paper page 70; original question and option regions

Area of the region (in sq. units) bounded by the curve $y = x^2 - 5x + 4$, $x = 0$, $x = 2$ and the X-axis is

వక్రము $y = x^2 - 5x + 4$, $x = 0$, $x = 2$ మరియు X-అక్షములచే పరిబద్ధమైన ప్రాంతం యొక్క వైశాల్యం (చ॥ యూనిట్లలో)

Options :

$$\frac{8}{3}$$

1. ✖

$$3$$

2. ✔

$$5$$

3. ✖

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Maths - Question 77 (continued)

Original paper page 71; original question and option regions

$$\frac{5}{2}$$

Maths - Question 78

Original paper page 71; original question and option regions

If the order and degree of the differential equation $x \frac{d^2y}{dx^2} = \left(1 + \left(\frac{d^2y}{dx^2}\right)^2\right)^{-1/2}$ are

k and l respectively, then k, l are the roots of

$$x \frac{d^2y}{dx^2} = \left(1 + \left(\frac{d^2y}{dx^2}\right)^2\right)^{-1/2}$$

అవకలన సమీకరణము యొక్క పరిమాణము, తరగతులు k మరియు

l లు అయితే, k, l లు మూలాలుగా గల సమీకరణం

Options :

$$x^2 - 5x + 6 = 0$$

1. ✖

$$x^2 - 3x + 2 = 0$$

2. ✖

$$x^2 - 7x + 12 = 0$$

3. ✖

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Maths - Question 78 (continued)

Original paper page 72; original question and option regions

$$x^2 - 6x + 8 = 0$$

Maths - Question 79

Original paper page 72; original question and option regions

The equation of the curve passing through the point $(0, \pi)$ and satisfying the differential equation $ydx = (x + y^3 \cos y)dy$ is

$(0, \pi)$ బిందువు గుండాపోతూ, $ydx = (x + y^3 \cos y)dy$ అనే అవకలన సమీకరణాన్ని తృప్తిపరిచే వక్రము యొక్క సమీకరణం

Options :

1. ✖ $x = y^2 \sin y + y \cos^2 y$

2. ✔ $x = y^2 \sin y + 2y \cos^2 \frac{y}{2}$

3. ✖ $x = y^2 \sin y + y \cos^2 \frac{y}{2}$

4. ✖ $x = y^2 \sin y - y \cos^2 y$

Maths - Question 1

Original paper page 2; original question and option regions

$$\text{If } A = \left\{ x \in \mathbb{R} / \sin^{-1} \left(\sqrt{x^2 + x + 1} \right) \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] \right\} \text{ and}$$

$$B = \left\{ y \in \mathbb{R} / y = \sin^{-1} \left(\sqrt{x^2 + x + 1} \right), x \in A \right\} \text{ then}$$

$$\text{If } A = \left\{ x \in \mathbb{R} / \sin^{-1} \left(\sqrt{x^2 + x + 1} \right) \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] \right\} \text{ మరియు}$$

$$B = \left\{ y \in \mathbb{R} / y = \sin^{-1} \left(\sqrt{x^2 + x + 1} \right), x \in A \right\} \text{ అయితే, అప్పుడు}$$

Maths - Question 1 (continued)

Original paper page 3; original question and option regions

$$A \cap B \neq \phi$$

$$A \cap B^C = [0, 1]$$

$$A^C \cap B = \left[\frac{\pi}{3}, \frac{\pi}{2} \right]$$

$$A \cup B = \mathbb{R} - \left\{ [-1, 0] \cup \left[\frac{\pi}{3}, \frac{\pi}{2} \right] \right\}$$

Maths - Question 2

Original paper page 3; original question and option regions

The domain of the function $f(x) = \sqrt{\log_e \left(\frac{1}{x^2 - 4x + 4} \right)} + \sin^{-1}(x^2 - 2)$ is

$$f(x) = \sqrt{\log_e \left(\frac{1}{x^2 - 4x + 4} \right)} + \sin^{-1}(x^2 - 2) \text{ ప్రమేయం యొక్క ప్రదేశము}$$

Maths - Question 2 (continued)

Original paper page 4; original question and option regions

$$[1, 3]$$

$$[1, 3)$$

$$[1, \sqrt{3}]$$

$$[1, \sqrt{3})$$

Maths - Question 3

Original paper page 4; original question and option regions

For all $n \in \mathbb{N}$, if $n(n^2 + 3)$ is divisible by k , then the maximum value of k is

అన్ని $n \in \mathbb{N}$ లకు, $n(n^2 + 3)$ అనేది k చే భాగించబడితే, k యొక్క గరిష్ఠ విలువ

Options :

1. ✖ 4

2. ✖ 6

3. ✖ 8

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Maths - Question 3 (continued)

Original paper page 5; original question and option regions

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Maths - Question 4

Original paper page 5; original question and option regions

If a is the determinant of the adjoint of the matrix $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 3 \end{bmatrix}$ and b is the

determinant of the inverse of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & -3 & -1 \\ 2 & 1 & -4 \end{bmatrix}$ then $\frac{b+1}{18b} =$

$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 3 \end{bmatrix}$ మాత్రిక యొక్క అనుబంధ మాత్రికకు నిర్ధారకం a మరియు $\begin{bmatrix} 1 & 2 & 3 \\ 4 & -3 & -1 \\ 2 & 1 & -4 \end{bmatrix}$ మాత్రిక

యొక్క విలోమానికి నిర్ధారకం b అయితే $\frac{b+1}{18b} =$

Options :

1. ✓ a

Maths - Question 4 (continued)

Original paper page 6; original question and option regions

$$10a$$

$$2 + a$$

$$2a$$

Maths - Question 5

Original paper page 6; original question and option regions

Consider two systems of 3 linear equations in 3 unknowns $AX = B$ and $CX = D$. If $AX = B$ has unique solution D and $CX = D$ has unique solution B , then the solution of $(A - C^{-1})X = O$ is

3 అజ్ఞాత రాశులలో మూడు ఏక ఘాత సమీకరణాలను గలిగిన రెండు వ్యవస్థలు $AX = B$ మరియు $CX = D$ లను పరిగణించండి. $AX = B$ ఏకైక సాధన D ని మరియు $CX = D$ ఏకైక సాధన B ని కలిగి ఉంటే, $(A - C^{-1})X = O$ యొక్క సాధన

Options :

1. ✖ B

2. ✔ D

3. ✖ B + D

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Maths - Question 5 (continued)

Original paper page 7; original question and option regions

B – D

Maths - Question 6

Original paper page 7; original question and option regions

$f(x)$ is an n^{th} degree polynomial satisfying $f(x) = \frac{1}{2} \begin{vmatrix} f(x) & f\left(\frac{1}{x}\right) - f(x) \\ 1 & f\left(\frac{1}{x}\right) \end{vmatrix}$. If

$f(2) = 33$, then the value of $f(3)$ is

$f(x)$ అనేది $f(x) = \frac{1}{2} \begin{vmatrix} f(x) & f\left(\frac{1}{x}\right) - f(x) \\ 1 & f\left(\frac{1}{x}\right) \end{vmatrix}$ ని తృప్తిపరచే ఒక n వ తరగతి బహుపది. $f(2) = 33$

అయితే, $f(3)$ యొక్క విలువ

Options :

126

1. ✖

214

2. ✖

Maths - Question 6 (continued)

Original paper page 8; original question and option regions

244

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